

Turning Your Wind Instrument into a Music Controller: Real-time Fingering Estimation by Classifying Reflected White Noise

Naoyuki Terashita
Independent
Tokyo, Japan

Kyoichi Sugahara
Independent
Tokyo, Japan

Yosuke Terashita
Independent
Tokyo, Japan

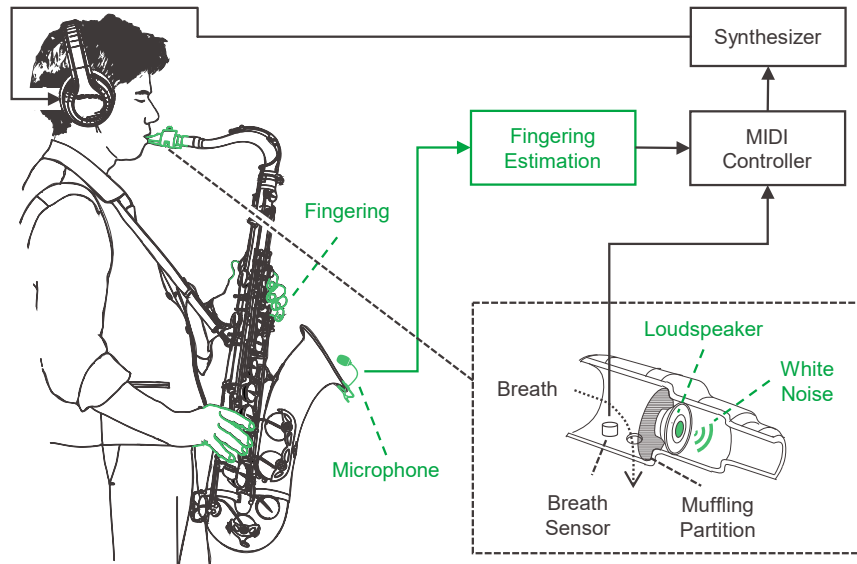


Figure 1: Illustration of the concept of a wind controller incorporating our fingering estimation method. The green lines highlight the scope of this paper, i.e., our fingering estimation method. The loudspeaker plays white noise continuously, and fingerings are estimated from the acoustic characteristics of sound pressures recorded at the end of the wind instrument. Our method is intended to be combined with a breath sensor to measure wind speed and the timing of breaths. A synthesizer generates sound signals on the basis of estimated fingering and detected breath and feeds them back to the player.

ABSTRACT

We propose a novel fingering estimation method that allows a player to use any wind instrument as a music controller by attaching a single microphone and loudspeaker-embedded mouthpiece. The loudspeaker plays white noise continuously while the fingerings are estimated in real-time based on the sound pressure recorded at the end of the instrument. Our method addresses a major problem of conventional music controllers: differences in the tactile feel of keys compared to the player's own instrument. We demonstrated that the proposed method accurately estimated fingerings on a

saxophone with promising performance (a 1.05 % misclassification rate), satisfying the low-latency feedback required in the context of musical performance.

CCS CONCEPTS

• Human-centered computing → Interaction techniques; • Applied computing → Sound and music computing.

KEYWORDS

Music; Acoustics; Wind Instruments; Latency; Estimation; Fingering

ACM Reference Format:

Naoyuki Terashita, Kyoichi Sugahara, and Yosuke Terashita. 2021. Turning Your Wind Instrument into a Music Controller: Real-time Fingering Estimation by Classifying Reflected White Noise. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '21 Extended Abstracts)*, May 8–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3411763.3451719>

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

CHI '21 Extended Abstracts, May 8–13, 2021, Yokohama, Japan

© 2021 Association for Computing Machinery.

ACM ISBN 978-1-4503-8095-9/21/05...\$15.00

<https://doi.org/10.1145/3411763.3451719>

1 INTRODUCTION

Several studies have reported that playing musical instruments has a positive impact on individuals' health and abilities [2, 7]. However, not all musicians can experience these benefits due to a lack of resources (e.g., transportation, time, and money) for dedicated practice spaces, such as studios. One reasonable way to compensate for a lack of such resources is to practice at home using music controllers. For example, a piano player can practice finger movements with a music controller that simulates the tactile experience of a piano (i.e., a MIDI keyboard). Music controllers are typically connected to synthesizers to generate audio signals. A player can listen to the synthesized sound feedback through headphones, suppressing the output sound so that it does not exceed the acceptable level of their living environment. This paper focuses on a music controller for wind instruments, commonly termed a *wind controller*. Despite the large number of wind instrument players, interfaces of wind controllers have been less thoroughly studied than those of other instruments, such as the piano and drums.

A wind controller typically consists of two components, a breath sensor and fingering detection sensor, to determine sound intensity and the notes to play. Hereinafter, we refer to the combination of open and closed holes on a wind instrument as *fingering*. We argue that current interfaces for fingering detection are not suitable for practice due to the difference in tactile sensations of the keys compared to the player's actual instrument. In the most common form of wind controller, fingerings are detected using physical sensors [1, 17, 26]. These physical sensor-based controllers are reported to cause frequent playing errors [21]. Other studies [14, 15] have proposed using actual wind instruments with sensor-embedded keys, which allow players to experience real key sensations. However, practicing on sensor-embedded instruments does not always improve player's ability on their instrument, since every wind instrument has different key-weight characteristics.

To this end, we propose a novel fingering detection method that allows players to use their own instruments as wind controllers. Our method only requires a player to attach a microphone and mouthpiece with a built-in loudspeaker to the player's saxophone (Figure 1). The loudspeaker in the mouthpiece plays white noise continuously, and the fingering is estimated based on the sound pressure reflected in the body and recorded at the end of the wind instrument. Our method also enjoys the favorable property of reasonable economic cost compared to other similar methods.

We conducted experiments with an actual saxophone to evaluate our method. We confirmed that our method precisely inferred true fingerings while satisfying low-latency and low-sound-level requirements.

Our main contributions can be summarized as follows:

- We propose a novel fingering detection method for wind instruments that estimates player's fingering on the basis of how the low-level white noise played in the mouthpiece is deformed through the body of the instrument.
- We demonstrate that the true fingerings on an actual wind instrument are precisely inferred, confirming that the performance of this device is promising for the purpose of constructing a music controller.

2 RELATED WORK

2.1 Finger Actions in Wind Instruments

Several prior works have studied the importance of improving finger actions in playing performance and indicate the effectiveness of practicing finger actions on the player's own instrument.

Wind instrument players control multiple parameters to produce desired tones. These parameters include finger forces, breath pressures, and forces with the lips [24]. Among these parameters, a prior study with saxophonists [8] suggests that the finger actions are especially important for timing in saxophone performance.

We also emphasize that the finger actions should be practiced on the player's own instrument. Clarinet teachers recommend using only minimal forces with fingers to close tone holes [23]. These minimal forces depend on the particular characteristics of the levers and springs in the keys of each instrument. Thus, minimal finger forces learned in playing an instrument other than the player's instrument are not always optimal for the player's own instrument.

2.2 Existing Fingering Detection Approaches

The most common type of wind controller is what we term a hardware-based wind controller, which simulates a mouthpiece, keys, and instrument body with a breath controller, sensors, and plastic case, respectively. For example, [17, 26] use physical buttons and [1] uses touch sensors to detect player's fingerings. However, the keys of hardware-based wind controllers provide very different tactile sensations compared to actual instruments. [21] reported that it takes a long time for saxophonists to adapt to hardware-based wind controllers, resulting in more frequent playing errors than in playing actual saxophones.

Other works [6, 14, 15, 19] propose wind instruments that embed sensors around keys to detect fingerings, which we term sensor-embedded wind controllers. [14, 15, 19] use hall effect sensors and [6] uses capacitance sensors for the fingering detection. Unlike hardware-based controllers, these controllers use real instruments, allowing players to experience real key-action sensations. However, they require that a wind instrument other than the player's instrument is prepared to construct the sensor-embedded wind controller. Additionally, it is not practical to attach the sensors to a player's instrument non-invasively since straightforward implementation requires that as many sensors and wires as there are keys be placed carefully so they do not interfere with the finger movements.

Unlike those approaches, our method allows players to use their own instruments as wind controllers with two easily detachable devices: a microphone and a loudspeaker-embedded mouthpiece. Thus, players are expected to be able to improve their ability of finger actions on their instruments more effectively.

2.3 Practice Mutes

[3, 16, 18] propose the use of practice mutes for saxophones that cover the entire instrument with a muffling bag. Although these approaches are well suited to our aim to facilitate practicing on a player's own instrument at home, we argue that they often suffer from expensive costs and limited playability. Since muffling bags need to be designed to fit the distinctive shape of each instrument (e.g., alto, tenor, soprano saxophones, and clarinets), players must

purchase different practice mutes for each of their instruments. In addition, since better muting performance requires larger spaces for muffling, users have to manage a trade-off between muting ability and freedom of arm and hand positions when playing the instrument. The wind controller combined with our fingering detection method is expected to provide unrestricted playability in terms of fingering at a reasonable cost.

2.4 Interfaces with acoustics

A general summary of our proposal is that it uses acoustic information to construct a tangible interface, which approach has been studied in several works [11, 13, 22]. [22] proposes a touch-sensing technique called Touch & Activate. It recognizes contextual aspects of touches, such as grasping, by attaching a vibration speaker and a piezo-electric microphone to any physical objects. Acoustrument [11] uses a speaker-microphone pathway structure and recognizes the user's input to the structure through the altered acoustic output. We recognize that these techniques are compatible with both our goal of non-invasive sensing and the nature of wind instruments, where each fingering has unique acoustic characteristics. Our method differs from these approaches in using white noise instead of their swept-frequency acoustic signals to satisfy the low-latency feedback ($\sim 10^{-2}$ sec) required in the context of musical performance.

3 PROPOSED METHOD

In Subsection 3.1, we first propose a formulation of the fingering estimation problem. Section 3.2 then introduces our algorithm that solves the formulated problem. Section 3.3 explains the latency requirement of the music controller, which limits the length of the sampling duration used in the proposed algorithm. We also discuss an expected usage of our method in wind controllers in Section 3.4.

3.1 Formulation of Fingering Estimation

We formulate the fingering estimation problem through a waveguide model of an instrument and its transfer functions. Several works [5, 20] have reported that a wind instrument can be modeled using a 1-dimensional waveguide model. The waveguide model takes sound pressure at the end of the mouthpiece as an input and sound pressure outside of the instrument as an output. Let K be the number of fingering candidates. Since each fingering $k \in \{1, \dots, K\}$ of the saxophone corresponds to a different musical note, each note provides a unique transfer function $H_k \in \{H_1, \dots, H_K\}$ of the waveguide model. Let $X(\omega)$ and $Y(\omega)$ be input sound pressure and output sound pressure, respectively, corresponding to an angular velocity $\omega \in \mathbb{R}$ in the frequency domain. $Y(\omega)$ in steady state is given by $Y(\omega) = H_k(\omega)X(\omega)$.

The effects of fingerings can be observed as changes in the amplitude of $Y(\omega)$. We denote amplitude by $|\cdot|$. When we use white noise for the input sound pressure, $X(\omega)$ has a constant amplitude $c \in \mathbb{R}$ over the entire range of frequencies, $|X(\omega)| = c$. Additionally, the amplitude of output sound pressure is given by $|Y(\omega)| = c|H_k(\omega)|$. We then obtain an amplitude spectrum vector of reflected sound

$\mathbf{y}(k)$ in the discrete frequency domain as follows:

$$\begin{aligned} \mathbf{y}(k) &:= [|Y(\omega_0)|, \dots, |Y(\omega_N)|] \\ &= c [|H_k(\omega_0)|, \dots, |H_k(\omega_N)|]. \end{aligned} \quad (1)$$

Let Δt_{samp} be window size. The frequency component is given by $\omega_n = 2\pi n / \Delta t_{\text{samp}}$, where $n \in \{0, 1, \dots, N\}$. The index of the largest frequency component N is given by $N = \Delta t_{\text{samp}} f_s / 2$ in the sampling theorem, where f_s is the sampling frequency. In this setting, $\mathbf{y}(k)$ is not dependent on input X and is determined only by the player's fingering k . We denote fingering estimation as a task that obtains an estimator $\hat{k}_\theta(\mathbf{y})$ with a parameter θ that effectively approximates the inverse function $\mathbf{y}(k)^{-1}$ to predict the fingering k .

3.2 Algorithm

Let us consider the estimation procedure of the player's fingering $k(t)$ at time t (Figure 2). The loudspeaker attached to the mouthpiece plays white noise continuously, and the microphone placed outside the body of the instrument records reflected sound pressure. Note that the noise level is set below an acceptable noise limitation for residences (~ 55 dB [9]), and so it does not conflict with the purpose of silencing. As soon as the reflected sound pressure between t and $t + \Delta t_{\text{samp}}$ is obtained, the amplitude spectrum vector $\mathbf{y}(k, t)$ is calculated using STFT. The predefined estimator $\hat{k}_\theta(\mathbf{y})$ outputs a predicted fingering: $\hat{k}_\theta(\mathbf{y}(k, t)) = k_{\text{pred}}(t) \in \{1, \dots, K\}$. The above process is performed at a regular interval of ΔT to track the player's fingerings on the instrument over time.

In this paper, we use a machine learning classification model for the estimator. Since determining the frequency response of the body of the wind instrument is complicated due to its mechanics (e.g., register-key), a single feature of the amplitude spectrum (e.g., fundamental frequency) alone does not provide sufficient information to classify a player's fingerings (see Appendix A.2 for a detailed discussion). We expect the machine learning model to learn to classify the fingerings according to fundamental frequencies as well as other aspects of the amplitude spectrum (e.g., *cut-off frequencies* and the amplitude of peaks weakened by register-keys [4]).

In our method, the goal of machine learning is to find parameters θ that provide an estimator $\hat{k}_\theta(\mathbf{y}) = k_{\text{pred}} \sim k$. We train the parameters θ a priori, using sets of a true fingering k' and the corresponding amplitude spectrum vector $\mathbf{y}(k')$.

3.3 Latency Requirement

One of the major technical challenges of the proposed method is to satisfy the latency requirement for the music controller. In this paper, latency $\Delta t_{\text{latency}}$ is the delay between time t and the time at which the synthesized sound based on $k_{\text{pred}}(t)$ is fed back to the player.

Prior works [10, 12] have revealed that latency does not harm players' performance as long as it is kept within a certain range, which we call the latency requirement. Although the latency requirement Δt_{max} is highly dependent on the context of the application [12], [10] reported that a latency up to 30.0 ms is acceptable for most musical applications.

Considering the latency requirement, the window size of STFT Δt_{samp} should be carefully chosen; a larger window size Δt_{samp}

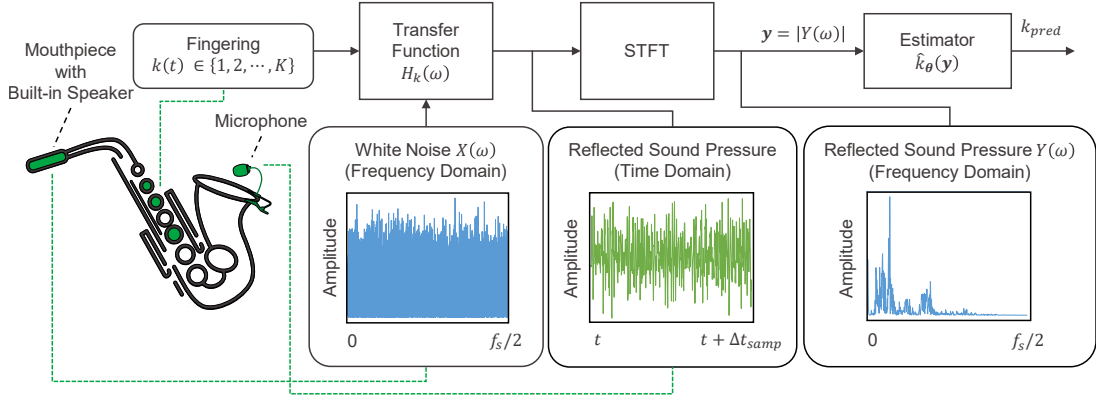


Figure 2: Overview of the fingering estimation method. We assume each fingering $k(t)$ leads to its unique transfer function $H_k(\omega)$, which deforms the frequency characteristics of white noise $X(\omega)$ into output sound pressure $Y(\omega)$. The discrete frequency spectrum $y(k, t)$ is obtained by Short-Time Fourier Transform (STFT). Finally, the estimator $\hat{k}_\theta(y) = k_{pred}$ predicts the fingering $k(t)$.

directly increases the latency, and a small Δt_{samp} leads to poor frequency resolution. In the experiment in Section 4, we adopt the window size $\Delta t_{samp} = 25.0$ ms, considering the latency of 30.0 ms [10] to be acceptable and the post-processing after the sampling to take less than 5.0 ms. The post-processing includes SFTF, fingering estimation, and synthesizing sound signals. A detailed discussion and preliminary experiments for determining the duration of the post-processing are included in Appendix A.1.

3.4 Use Cases

Although the construction of a complete wind controller is beyond the main focus of this paper, we provide an expected usage of our method and indicate that a complete controller will not lose the advantage of non-invasive sensing and easy detachment.

We expect that the complete wind controller will be designed so that the player's breath passes through an air path separate from that of the white noise, and the sound intensity and timing are determined based on the output value of a breath sensor installed in spatial separation from the loudspeaker (Figure 1). For example, the breath sensor can be separated from the loudspeaker by a muffling partition in the mouthpiece so that the noise from the player's breathing does not affect the sound recorded by the microphone.

It is also possible to attach a reed-shaped pressure sensor, which is used in [17, 26], to the mouthpiece in order to allow the player to control the pitch without producing actual sounds from the reed.

4 EXPERIMENT: CLASSIFICATION PERFORMANCE

To verify that our method can accurately estimate the player's fingerings in a real-time scenario, we evaluated fingering classification performance using an actual saxophone.

4.1 Setup

We used an alto saxophone (Yamaha YAS-34II) and fingering candidates of $K = 32$ as described in [27]. These fingerings cover the

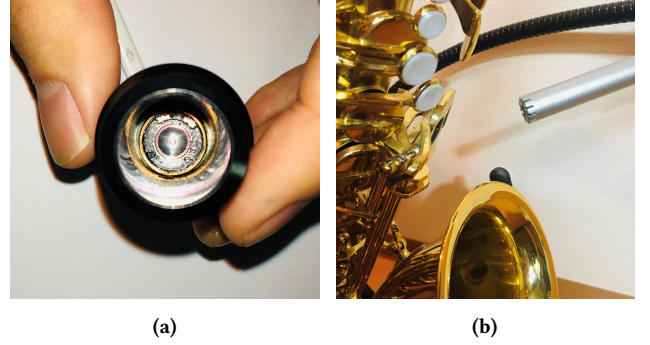


Figure 3: Hardware setup of the experiment. A loudspeaker (a) is embedded in the mouthpiece and a condenser microphone (b) is supported by an arm stand.

common alto saxophone range of written A $\sharp$ 3 to F6. A driver unit from a pair of earbuds (Apple EarPods) was embedded in the mouthpiece for the loudspeaker (Figure 3a), and a condenser microphone (Behringer ECM8000) was placed approximately 10 cm above the bell (Figure 3b). We used a USB audio interface (Yamaha Steinberg CI1) with a sampling rate $f_s = 48.0$ kHz.

We collected sets of true fingerings k' and the corresponding amplitude spectrum vectors $y(k')$ to train the estimator $\hat{k}_\theta(y)$. We played white noise in the mouthpiece and recorded the sound pressure of each fingering k' separately for 30.0 sec. We then applied STFT with the window duration Δt_{samp} . By shifting the window position with the interval $\Delta T = 10.0$ ms, we constructed a set of amplitude spectrum vectors $y(k')$.

For the test data, we collected sound pressures independently from the training data. To ensure the correspondence between the true fingerings $k(t)$ and amplitude spectrum vectors $y(k, t)$, we separately recorded the sound pressure of each fingering for 10.0 sec and then concatenated them in random order. This resulted in a reflected noise sound with the length $T = 10.0 \times 32 = 320.0$ sec.

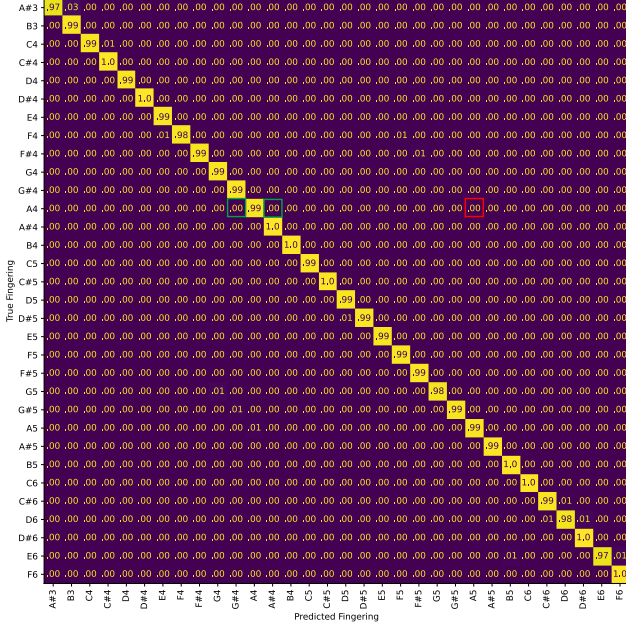


Figure 4: Confusion matrices between true fingerings (vertical) and estimated fingerings (horizontal) from Section 4. Examples of error patterns in A4 samples are visualized with an octave error in the red box and half-tone errors in green boxes of A4 samples.

We evaluated the performance of the fingering estimation following the algorithm in Section 3.2. Starting from $t = 0$ ms in the test data, we calculated $\mathbf{y}(k, t)$ with STFT and then calculated $\hat{k}_\theta(\mathbf{y}(k, t)) = k_{pred}(t)$. By iteratively adding $\Delta T = 5.0$ ms to t unless $t + \Delta T \leq T$, we obtained a series of predictions $k_{pred}(t)$. Finally, we calculated the misclassification rate using the sets of $k(t)$ and $k_{pred}(t)$.

4.2 Results and Discussion

Figure 4 shows the confusion matrix obtained from true and predicted fingerings. The overall misclassification rate was 1.05 %. The performance of our method seems to be promising for the purpose of the fingering detection in wind controllers.

We further investigated the sources of misclassification to understand the characteristics of our method. We categorised the misclassification cases into three error patterns: *octave*, *half-tone*, and other errors. An octave error is defined as an instance in which a sample was incorrectly classified as a note that is an octave higher or lower than the true note, such as an A4 sample classified as A5 (the red box in Figure 4). We classify a misclassified sample as a half-tone error when the predicted note is a half-tone higher or lower than the true note—such as A4 samples classified as G#4 or A#4 (the green boxes in Figure 4). Table 1 shows the percentage of the octave and half-tone errors in the total misclassified samples.

The most frequently encountered type of error was half-tone errors. This is reasonable since the window size of 25.0 ms only provides the frequency resolution $\Delta f_{samp} = 1/(25.0 \times 10^{-3}) =$

Table 1: Percentages of the two error patterns, octave and half-tone errors, in the misclassified samples. To save space, we only present the results of notes with error patterns greater than 1.5 % (bold values).

Note	Octave [%]	Half-tone [%]
A#3	0.00	9.82
C4	0.00	2.53
F4	1.79	2.23
F#4	3.42	0.15
D5	0.45	1.79
D#5	0.60	2.23
G5	4.17	0.60
G#5	2.68	0.74
A5	2.53	0.15
C#6	0.89	1.79
D6	0.00	5.95
E6	0.15	3.72
Other	13.54	11.46
Total	30.21	43.15

40.0 Hz, which is insufficient compared to the frequency interval between adjacent notes (e.g., the difference in fundamental frequencies between written A4 and A#4 of the alto saxophone is $26.2 \text{ Hz} < 40.0 \text{ Hz} = \Delta f_{samp}$. In this view, we suppose that the low misclassification rate of 1.05 % is achieved with the help of the information in higher frequency range (e.g., peak frequencies of sub-harmonics).

Another interesting observation is that most of the octave errors occur between pairs of the notes that fingerings differ only in the presence or absence of the register-key (highlighted octave errors in Table 1). This seems to be originated from the nature of those pairs of fingerings; they share the same fundamental frequencies [4]. Frequent octave errors do not observed in the samples of notes that are not combined with the register-key (e.g., D6~F6).

5 LIMITATIONS AND FUTURE WORK

5.1 Practical Challenges of Designing a Music Controller

To build a music controller that leverages our fingering estimation method, we must further examine the practical requirements considering user experiences. We consider the accuracy of our approach at this point to be possibly insufficient for practical use since a player cannot tell whether a wrong note is caused by their mistake or a misclassification of the fingering. Further evaluation of the proposed method should thus incorporate user studies conducted in musical contexts. We believe the evaluation of fingering estimation and improving estimation performance are promising directions for future research.

5.2 Applicability to Other Wind Instruments

Our method assumes that each fingering corresponds to a unique transfer function of the instrument. As long as this assumption holds, our fingering estimation method should be compatible with

other wind instruments besides the saxophone, such as the clarinet, oboe, and flute. These wind instruments also have separate mouthpieces that can be replaced with our loudspeaker-embedded mouthpieces.

5.3 Detecting Overblowing

In several wind instruments, including the flute and piccolo, fingerings alone cannot determine the notes the musician intends to play. This is because players often control the notes they play through an *overblowing* technique. For example, the flute's fingering of E4 is identical to that of E5, and flutists control the octave by changing the speed and shape of the air they blow into the instrument.

Thus, the development of a wind controller for the flute must incorporate both fingering detection and breath sensing to determine the note a flutist intends to play. The problem of fingering alone not determining the note is not only the case for our method but has also appeared in other fingering detection approaches.

6 CONCLUSION

We have proposed a novel fingering estimation method that enables a player to use their own wind instrument as a music controller by attaching a single microphone and mouthpiece with a built-in loudspeaker. The loudspeaker in the mouthpiece plays white noise continuously, and the fingering is estimated from the reflected sound pressure recorded at the end of the wind instrument. We evaluated the accuracy of the estimated fingerings using the actual saxophone. We observed a misclassification rate of 1.05 % confirming that the performance of our method is promising for the purpose of the fingering detection in music controllers.

A APPENDICES

A.1 Determining Window Size

Sources of the Latency. Let $\Delta t_{latency} := \Delta t_{samp} + \Delta t_{post}$, where Δt_{post} is the post-processing duration after the sampling. Since longer Δt_{samp} gives a better frequency resolution ($= 1/\Delta t_{samp}$), Δt_{samp} should be as long as possible for better estimation performance. Thus, we determined Δt_{samp} by $\Delta t_{samp} = \Delta t_{max} - \Delta t_{post}$.

We further decomposed the source of the post-processing duration Δt_{post} . Let $\Delta t_{post} := \Delta t_{STFT} + \Delta t_{est} + \Delta t_{synth}$, where Δt_{STFT} , Δt_{est} , and Δt_{synth} are the processing duration for STFT, the fingering estimation, and synthesizing sound signals, respectively. As prior work [25] reported, we assume that Δt_{synth} is ~ 4.0 ms. For the remaining sources of latency, we experimentally evaluated Δt_{STFT} and Δt_{est} .

Experimental Setup. We measured the duration of Δt_{STFT} and Δt_{est} . We adopted STFT with a Hamming window for Δt_{STFT} and multinomial logistic regression as the estimator $\hat{k}_\theta$ for Δt_{est} . The output dimension of the estimator was set as $K = 32$, which is the common number of fingering candidates for the saxophone. Since their computational costs depend on the number of frequency components $N = \Delta t_{samp} f_s / 2$, we measured Δt_{STFT} and Δt_{est} by changing Δt_{samp} . We implemented a series of STFT and multinomial logistic regression processes using MATLAB 2020a and measured the average processing duration of 1000 iterative executions by changing Δt_{samp} . The PC used for the calculation had

a Intel Core i7-4790K processor, with a maximum frequency of 4.00 GHz and 16.0 GB of memory.

Results. We found that both Δt_{STFT} and Δt_{est} are negligibly small compared to the acceptable latency requirement Δt_{max} . For example, the average of the total processing duration was 1.37×10^{-2} ms when $\Delta t_{samp} = 25.0$ ms. From this result and $\Delta t_{synth} \approx 4.0$ ms, we concluded that Δt_{post} is smaller than 5.0 ms. Finally, we adopted $\Delta t_{samp} = 25.0$ ms, which satisfies the requirement that $\Delta t_{latency} = \Delta t_{samp} + \Delta t_{post} \leq \Delta t_{max} = 30.0$ ms

A.2 Fingerings and Acoustic Characteristics

Fingerings affect the frequency characteristics in the body of the wind instrument in a complicated way. Although the main role of fingering is to alter the effective length of the body to control the fundamental frequency, it also has other effects, including *register-keys* and *cut-off frequencies*.

Saxophone players can change the octave of a note by holding or releasing the register-key. Unlike other fingerings that control the note by changing the effective pipe length, the register-key does not change the effective length but controls the note by weakening the impedance of the fundamental frequency [4]. In other words, some pairs of octave different fingerings share the same fundamental frequencies, and the difference between them can be observed in the amplitude of the fundamental frequency.

Another effect of changes in the fingering is so-called cut-off frequencies. Cut-off frequencies are the specific frequencies excited by the first open tone hole below the lowest closed tone hole. Although low-frequency waves are reflected at the first open tone hole, higher frequency waves travel over the opened tone hole and sufficiently high-frequency waves will reach the end of the pipe over the opened holes [4]. Due to these high-frequency waves, the impedance spectra become irregular in a high-frequency range (around 800 Hz for the tenor and 1300 Hz for the soprano saxophone).

In summary, the low-frequency spectrum has important information, the fundamental frequency, which explicitly indicates the note, but it is not sufficient to classify the presence or absence of a register-key. On the other hand, the higher frequency spectrum has information of fingering as a form of cutoff frequency, but the relationship between fingering and cutoff frequency is not clear. Estimating fingering based on prior knowledge using cut-off frequencies is therefore difficult. Hence, we choose a machine learning model that can learn the implicit relationship between frequency response and fingering, and feed the entire band of frequency spectrum to the model to utilize all the information that is possibly helpful in estimating the fingerings.

REFERENCES

- [1] Akai. 2014. *EWI5000*.
- [2] Lars Ole Bonde, Knud Juel, and Ola Ekholm. 2018. Associations between music and health-related outcomes in adult non-musicians, amateur musicians and professional musicians—Results from a nationwide Danish study. *Nordic Journal of Music Therapy* 27, 4 (2018), 262–282.
- [3] Best Brass. 2005. *e-Sax Alto*.
- [4] Jer-Ming Chen, John Smith, and Joe Wolfe. 2009. Saxophone acoustics: introducing a compendium of impedance and sound spectra. *Acoustics Australia* 37, 1-19 (2009).

- [5] Paul Dickens, Ryan France, John Smith, and Joe Wolfe. 2007. Clarinet acoustics: introducing a compendium of impedance and sound spectra. *Acoustics Australia* 35, 1 (2007), 17.
- [6] Cristopher Ramos Flores, Jim W Murphy, and Michael Norris. 2019. HypeSax: Saxophone acoustic augmentation.. In *NIME*. 365–370.
- [7] Marie Forgeard, Ellen Winner, Andrea Norton, and Gottfried Schlaug. 2008. Practicing a musical instrument in childhood is associated with enhanced verbal ability and nonverbal reasoning. *PLoS one* 3, 10 (2008), e3566.
- [8] Alex Hofmann and Werner Goebel. 2014. Production and perception of legato, portato, and staccato articulation in saxophone playing. *Frontiers in psychology* 5 (2014), 690.
- [9] Gavin King, Marek Roland-Mieszkowski, Timothy Jason, and Daniel Rainham. 2012. Noise Levels Associated with Urban Land Use. *Journal of urban health : bulletin of the New York Academy of Medicine* 89 (06 2012).
- [10] Nelson Posse Lago and Fabio Kon. 2004. The quest for low latency. In *ICMC*.
- [11] Gierad Laput, Eric Brockmeyer, Scott E Hudson, and Chris Harrison. 2015. Acoustics: Passive, acoustically-driven, interactive controls for handheld devices. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*. 2161–2170.
- [12] Andrew P McPherson, Robert H Jack, Giulio Moro, et al. 2016. Action-sound latency: Are our tools fast enough? (2016).
- [13] Makoto Ono, Buntarou Shizuki, and Jiro Tanaka. 2013. Touch & activate: adding interactivity to existing objects using active acoustic sensing. In *Proceedings of the 26th annual ACM symposium on User interface software and technology*. 31–40.
- [14] Naoyuki Onozawa. 2011. Hybrid wind musical instrument and electric system incorporated therein. *The Journal of the Acoustical Society of America* 129, 6 (2011), 4100.
- [15] William R Perkins. 1982. Musical apparatus. US Patent 4,342,244.
- [16] David George Robinson. 2001. Saxophone muffler. US Patent 6,239,338.
- [17] Roland. 2016. *Aerophone AE-10*.
- [18] Saxmute.ONE. 2019. *Saxmute ONE/3*.
- [19] Sébastien Schiesser and Jan C Schacher. 2012. SABRe: The Augmented Bass Clarinet.. In *NIME*.
- [20] Tamara Smyth and Marjan Rouhipour. 2013. Saxophone modelling and system identification. In *Proceedings of Meetings on Acoustics ICA2013*, Vol. 19. Acoustical Society of America, 035010.
- [21] Laura A Stambaugh. 2015. Examination of a MIDI wind controller for use in wind performance research. *Music Performance Research* 7 (2015), 81–101.
- [22] Kentaro Takemura, Akihiro Ito, Jun Takamatsu, and Tsukasa Ogasawara. 2011. Active bone-conducted sound sensing for wearable interfaces. In *Proceedings of the 24th annual ACM symposium adjunct on User interface software and technology*. 53–54.
- [23] Reiner Wehle. 2007. *Clarinet Fundamentals Volume 2: Systematic Fingering Course, Clarinet Fundamentals*. Schott Music.
- [24] Joe Wolfe, André Almeida, Jer Ming Chen, David George, Noel Hanna, and John Smith. 2013. The player-wind instrument interaction. In *Proceedings of the Stockholm Music Acoustics Conference*. 323–330.
- [25] James Wright and Eli Brandt. 2001. System-level MIDI performance testing. In *ICMC*.
- [26] Yamaha. 1998. *Yamaha WX5 Wind MIDI Controller*.
- [27] Yamaha. 2020. How to Play the Saxophone:Saxophone fingering - Musical Instrument Guide - Yamaha Corporation. https://www.yamaha.com/en/musical_instrument_guide/saxophone/play/play002.html